

ALIBABA CLOUD AI HACKATHON PAKISTAN 2026

ParityLens

Catching bilingual exam errors before they reach students.

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THE PROBLEM

Two students. One exam. Two different questions.

Pakistani boards publish bilingual Urdu/English question papers so students can answer in the language they know best.

A documented Lahore-board case found Urdu and English versions of the same Chemistry and Physics questions that asked **genuinely different things** — not awkward wording, different questions.

This silently changes who gets which marks, based only on which language paper a student happened to read.

SOURCE: REPORTED LAHORE BOARD MISMATCH CASE

ENGLISH VERSION

"Which gas is **not** a greenhouse gas?"



URDU VERSION — ILLUSTRATIVE EXAMPLE

"کونسی گیس گرین ہاؤس گیس ہے؟"

("Which gas **is** a greenhouse gas?")

Illustrative example, not a real leaked question.

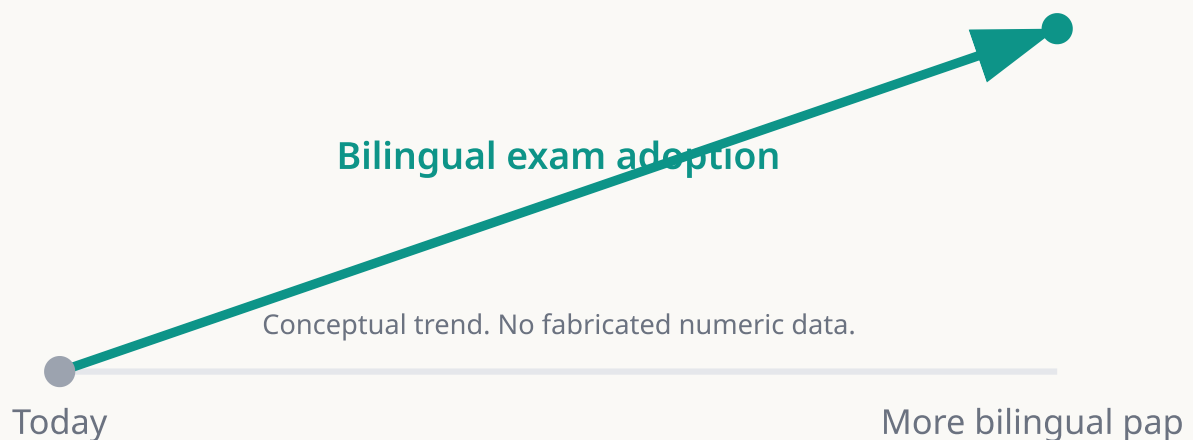
GROWING SURFACE AREA

This is expanding, not shrinking.

In 2026, the Punjab Board of Technical Education announced it will run Diploma of Information Technology exams in both Urdu and English — explicitly to give students a fairer chance to show what they know.

That is the right intention. But **every new bilingual paper is a new surface for silent drift**, and right now nothing checks this systematically at scale.

SOURCE: PUNJAB BOARD OF TECHNICAL EDUCATION, 2026 BILINGUAL DIT ANNOUNCEMENT



AI READINESS GAP

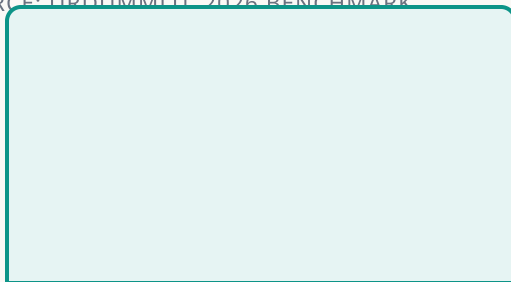
Urdu is still underserved by AI.

Generic translation or QA tools do not already solve this.

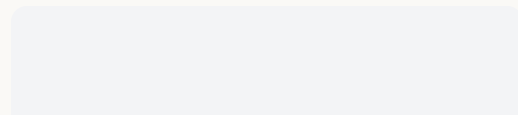
UrduMMLU (2026) is a benchmark built from 26,431 native Urdu multiple-choice questions drawn from real exam material. It found large performance gaps for current AI models on Urdu-language academic content.

That is exactly why a system for this language pair has to be purpose-built, not borrowed off the shelf.

SOURCE: URDUMMLU 2026 BENCHMARK



Performance gap on Urdu academic content



English academic content

Illustrative comparison, not exact benchmark scores.

PRODUCT DEFINITION

An independent QA gate that checks Urdu/English exam papers mean the same thing, before they print.

WHAT IT IS

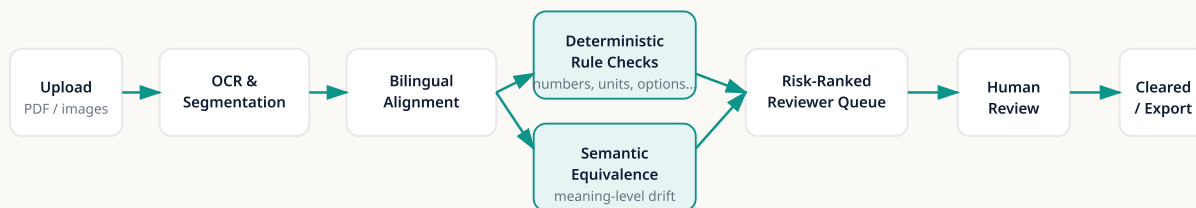
- ✓ An independent verification layer
- ✓ Applied after both language versions already exist
- ✓ Evidence-backed, with confidence scores
- ✓ Human-reviewed before any verdict

WHAT IT IS NOT

- × A translator
- × A question generator
- × An autograder
- × A fully automated rejection system

HOW IT WORKS

From upload to cleared paper.



Every flagged question is shown with the exact evidence that triggered it. A paper is only cleared after a human reviewer signs off.

MISMATCH TAXONOMY

Six ways a translation can quietly go wrong.

NUMERIC**Numeric mismatch**

"5 kg" becomes "50 kg"

UNIT**Unit mismatch**

kg vs g, °C vs °F

POLARITY**Negation / polarity**

Missing or added "not," "except," "least"

OPTIONS**Answer-option mismatch**

Options don't match in count or content

FORMULA**Formula / symbol mismatch**

H₂O vs CO₂, F = ma vs wrong symbol

ENTITY**Named-entity mismatch**

Newton vs Einstein, Lahore vs Karachi

CONDITION**Missing condition**

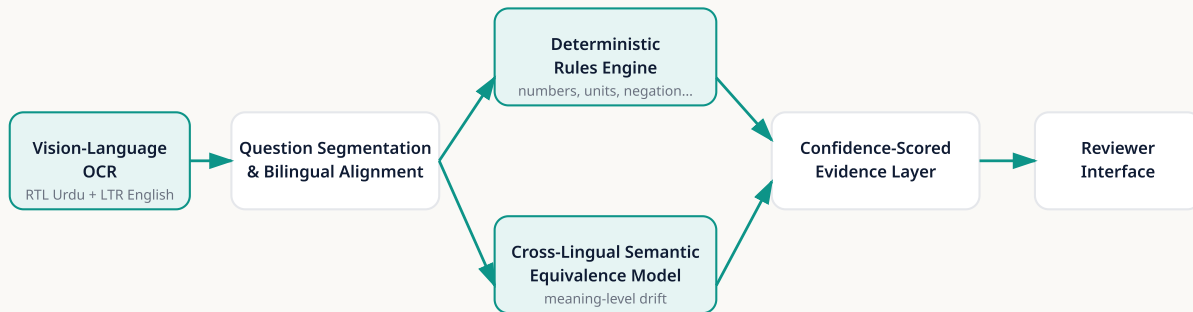
"at constant temperature" dropped

SEMANTIC**Semantic drift**

Both versions look fine, but ask different questions

TECHNICAL ARCHITECTURE

What's actually running under the hood.



Rules and semantic model run independently and are evaluated separately — a deliberate hybrid design.

REVIEWER EXPERIENCE

The human stays in control.

ENGLISH QUESTION

"A car travels **60 km** in **2 hours**. What is its average speed?"

URDU QUESTION

"ایک کار 60 کلومیٹر فی گھنٹہ چلتی ہے۔ اوسط رفتار کتنی ہے؟"

("60 km per hour" — unit/time condition differs)

**Unit / condition mismatch detected**

English: "in 2 hours" · Urdu: "per hour" · Confidence: 0.91

[Reject flag](#)[Accept flag](#)

A paper is only cleared after human sign-off. The system never declares a paper wrong on its own.

VALIDATION

How we're proving it actually works.

Benchmark of 100–300 real bilingual questions with controlled, seeded defects across every mismatch category.

- Precision, recall, F1 per category
- False positives per 100 questions
- End-to-end latency per paper

Three-way comparison: **rules-only** vs. **semantic-only** vs. **hybrid**.

Live benchmark results — insert after evaluation run

Placeholder for precision / recall / F1 numbers

DIFFERENTIATION

We are not competing with authoring tools.

The obvious question: don't multilingual assessment systems already exist?

Some do — for **authoring** and **translation**. ParityLens is neither. It is an independent **post-authoring verification gate** built around the specific failure modes of Urdu/English exam pairs.

As of an August 2026 public-web search, we did not find a close public Pakistan deployment matching this exact end-to-end workflow.

Honest framing

We are not claiming "world's first." We are claiming a narrow, defensible gap that no public system we could find is closing.

USE CASES

Anywhere a bilingual exam gets printed.

B**Examination boards**

BISE-style boards clearing SSC/HSSC and diploma papers before print.

T**Technical / diploma boards**

Bilingual diploma rollouts where numeric and unit drift is most likely.

S**Schools & coaching centers**

In-house bilingual test papers with no dedicated proofreading staff.

P**Publishers**

Bilingual practice material and past-paper compilations reproduced at scale.

G**Beyond Pakistan**

Any multilingual education system facing the same structural risk.

RESPONSIBLE AI

Built to be trusted with confidential exam material.

1

Human review is final

A paper is only cleared after a human reviewer signs off.

2

No automatic verdicts

Low-confidence flags are routed to a reviewer, never treated as fact.

3

Secure by default

Encrypted storage, short retention, role-based access.
Uploaded papers are not used to train models by default.

SCOPE DISCIPLINE

Deliberately narrow, on purpose.

Not in scope

- Student-facing app
- LMS integration
- Exam generator
- Plagiarism detector
- Full exam-board workflow replacement

Why this matters

Staying narrow is what made a working, evidence-backed prototype possible in six days. One job, done well.

Judges see a lot of broad ideas. A clearly bounded one is easier to believe.

CLOSING

ParityLens

ParityLens doesn't grade the exam. It makes sure both versions are asking the same one.

Next step: pilot with an examination board, school network, or publisher.